NaturArv3D Data Pipeline

Meeting 2026-10-06

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Introduction to `ome_zarr` CLI concept

- Have made PRs to `ome-zarr-py` repo, $2/4$ merged
  - TIFF stack export command: `ome-zarr-py#678`
  - Built-in support for `imagecodecs` codecs: `ome-zarr-py#679`
  - Even if not merged: shows how easily conversion tool can be made and maintained
- Reminder, importantly: many codecs are still not *standardised* in Zarr 3
  - Current proposal relies on one specific codec implementation (`imagecodecs`)
  - Upgrade path to standardised codec is expected, but not guaranteed
  - Proposed JPEG XL Extension Specification: `zarr-extensions#70`

Alternatives remain:

- DICOM with JPEG-LS or JPEG 2000 compression
- Uncompressed (perhaps losslessly compressed) TIFF stack

Demo `ome_zarr` CLI

1. Install `uv`:
   - *Some Python people may already have it installed*
     - *≈ 30% market share in October 2025: caveats apply, see source.*
     - *90.4K stars on GitHub (2026-10-05)*
   - Windows: `winget install --id=astral-sh.uv  -e`
   - Linux/MacOS: `curl -LsSf https://astral.sh/uv/install.sh | sh`
2. Install `ome-zarr-py` CLI:
   - `uv tool install --with imagecodecs-zarr git+https://github.com/JeppeKlitgaard/ome-zarr-py@feat-cli-convert`
3. Stream OME-Zarr and convert to TIFF stack (here we pick second level):
   - `ome_zarr export --format tiff-stack --level 2 --overwrite LINK_OR_LOC_OF_OME_ZARR output_tiffs`
   - Example:
     ```
     ome_zarr export --format tiff-stack --level 2 --overwrite "https://s3.g.s4.mega.io/aidjqnzhivz2k6kyxukiymjgfb4wxowhengdp/naturarv3d/CIL_basic_binned_2x2/NaturArv3D_format_viability_study/OME-Zarr/Canislupus_CZM_712__jpegxl_l_0_5__scales_2_4_8_16.ome.zarr" output_tiffs
     ```
4. *Optional: Open in Napari (`uvx --with pyqt6 napari output_tiffs`)*

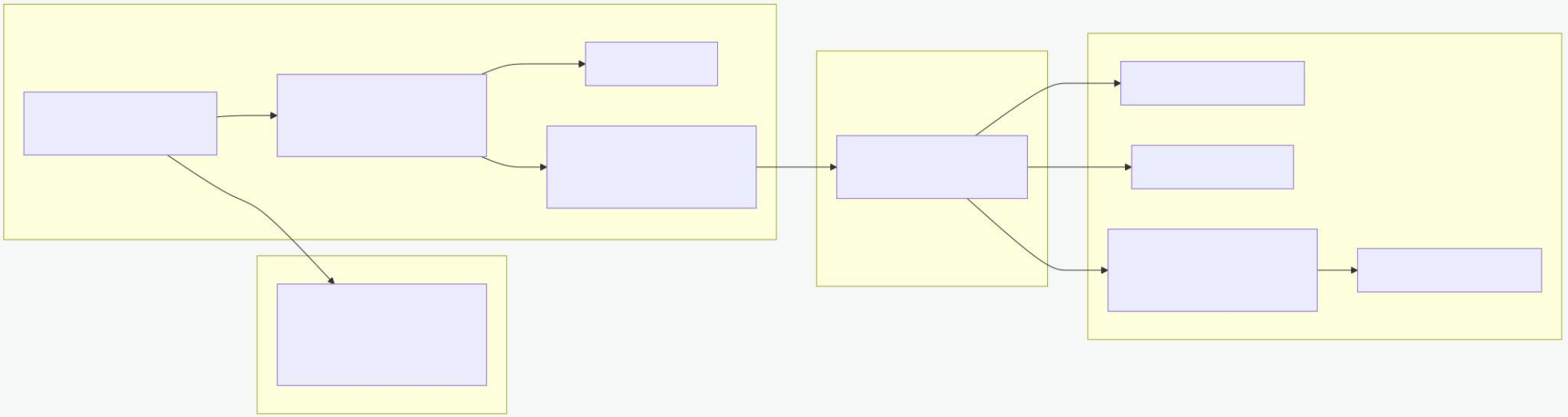
Tool Demo

Notes on tool

- Nice:
  - Streams in compressed format, so network friendly
  - Streams, so memory friendly
  - Emits standard TIFF stack
  - Preserves voxel size metadata
- Just a prototype
  - Intended to prove feasibility
  - Could just as well be a little graphical interface, or both

Data Flow Chart

- Very rough flow chart of data pipeline
- Mostly just included for later reference
  - *Exact timing of conversions and where compression happens is still TBD*.
  - Right-most half may be instructive

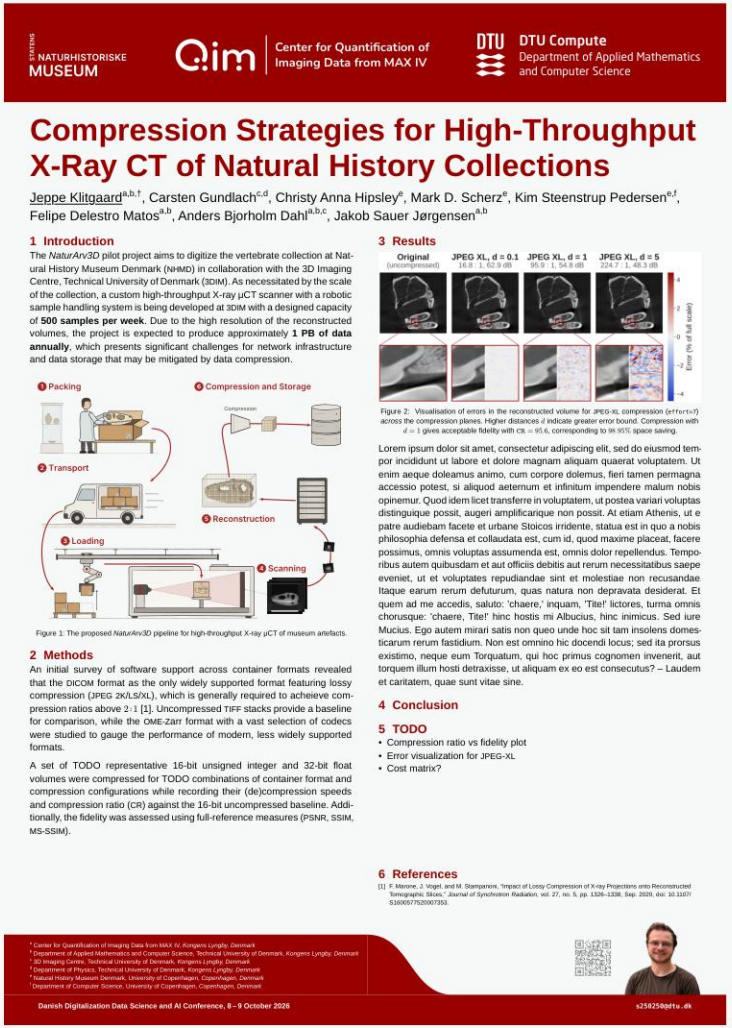


Summary

- Initially thought mostly about codecs
- Container format turned out to be more important
- May need to revisit codec when we get closer to implementation stage
- Don’t have to make decision now?

Aside: Poster at D3A 4.0 Conference (October 8 – 9)

- Bit last minute, *apologies*
- Hopefully will be engaging to people
- Non-finished version to the right
  - *(due to monotonicity of time)*



NATURHISTORISKE MUSEUM | Center for Quantification of Imaging Data from MAX IV | DTU Compute, Department of Applied Mathematics and Computer Science

Compression Strategies for High-Throughput X-Ray CT of Natural History Collections

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1 Introduction

The *NaturArv3D* pilot project aims to digitize the vertebrate collection at Natural History Museum Denmark (NHMD) in collaboration with the 3D Imaging Centre, Technical University of Denmark (3DIM). As necessitated by the scale of the collection, a custom high-throughput X-ray µCT scanner with a robotic sample handling system is being developed at 3DIM with a designed capacity of **500 samples per week**. Due to the high resolution of the reconstructed volumes, the project is expected to produce approximately **1 PB of data annually**, which presents significant challenges for network infrastructure and data storage that may be mitigated by data compression.

Figure 1: The proposed *NaturArv3D* pipeline for high-throughput X-ray µCT of museum artefacts.

2 Methods

An initial survey of software support across container formats revealed that the DICOM format as the only widely supported format featuring lossy compression (JPEG 2K/LS/XL), which is generally required to acheieve compression ratios above $2:1$ [1]. Uncompressed TIFF stacks provide a baseline for comparison, while the OME-Zarr format with a vast selection of codecs were studied to gauge the performance of modern, less widely supported formats.

A set of TODO representative 16-bit unsigned integer and 32-bit float volumes were compressed for TODO combinations of container format and compression configurations while recording their (de)compression speeds and compression ratio (CR) against the 16-bit uncompressed baseline. Additionally, the fidelity was assessed using full-reference measures (PSNR, SSIM, MS-SSIM).

3 Results

Figure 2: Visualisation of errors in the reconstructed volume for JPEG-XL compression (`effort=7`) across the compression planes. Higher distances d indicate greater error bound. Compression with $d = 1$ gives acceptable fidelity with $\mathrm{CR} = 95.6$, corresponding to 98.95% space saving.

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4 Conclusion

5 TODO

- Compression ratio vs fidelity plot
- Error visualization for JPEG-XL
- Cost matrix?

6 References

[1] F. Mannne, J. Vogel, and M. Stampanoni, "Impact of Lossy Compression of X-ray Projections onto Reconstructed Tomographic Slices," *Journal of Synchrotron Radiation*, vol. 27, no. 5, pp. 1326–1338, Sep. 2020, doi: 10.1107/S1600577520007353.

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Danish Digitalization Data Science and AI Conference, 8 – 9 October 2026

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Closing Remarks

- I may not have much more to contribute (at least for a bit?)
- Always happy to help, don’t hestitate to reach out
- *Everything (?)* available on private GitHub repo under `qim-center` org
  - https://github.com/qim-center/consult-xct-compression
    - QIM + Close friends already have access
    - Others can ask for inclusion
  - Slides and source code for every meeting presentation
  - Source code and outputs for all experiments and demos
    - Note: Obviously no volumetric data included
  - Poster including source code
  - Unfinished report that has been abandoned